MIL-C-24643/11B
3 October 1986
SUPERSEDING
MIL-C-0024643/11A(SH)
8 November 1985
MIL-C-24643/11
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSSSF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification not required.

Construction (non-watertight)

First - Copper conductor, AWG size 300 MCM, class G stranding, uncoated (see note 1).
Second - Separator.
Third - Ethylene propylene rubber insulation. Nominal wall thickness 0.075 inch.
Fourth - Reinforcement at manufacturer's option.
Fifth - Cross-linked polyolefin jacket bonded to the underlying insulation. Nominal wall thickness 0.075 inch.

NOTE: 1. The copper shall be hard drawn, having a minimum tensile strength of 67,200 lb/in^2, and minimum elongation of 0.85 percent in 60-inch lengths before stranding. The resistance specified is based on 102.7 percent of the values required for annealed copper.

TABLE I. Details.

Military part no. M24643/11	Type and size	Conductor size AWG	Overall diameter	
			minimum (inches)	maximum (inches)
-01UN	LSSSF-300	300 MCM (class G)	1.020	1.100

(B) denotes changes.

AMSC N/A FSC 6145
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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum..	0.0388
Voltage withstand - conductor to ground, volts root mean square, minimum.....	2500
Insulation resistance - megohms/1000 feet, minimum.....	100
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Cold bending, cable - at minus 30 ± 2°C, two turns around mandrel with diameter eight times that of specimen.....	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	700
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum.....	75
Elongation - percent of unaged, minimum..	75
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Heat distortion - percent of unaged, maximum.....	30

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Acid gas equivalent - percent, maximum	
Jacket	2
Fillers.....	--
Insulation.....	18

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	—
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	—
Insulation	1.5

UNIT ORDERING LENGTH:
Minimum 1045, maximum 1155 feet

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-11)

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